

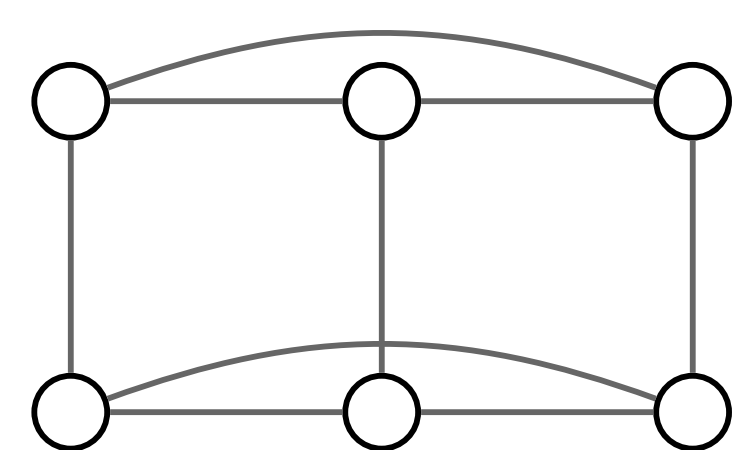
ABSTRACT

A set of vertices W **resolves** a graph G if each vertex in G is uniquely identified by its distance to every vertex in W . A **metric basis** is the minimum sized resolving set. A **reconfiguration graph** is a graph containing the solutions to a problem as vertices, where adjacency is determined by getting from one solution to another in one step. The **slide reconfiguration graph** acts under **token sliding** where two solutions are adjacent if one can be transformed into another by replacing a vertex with an adjacent vertex. We determine when the Cartesian product of two complete graphs has a connected slide graph.

METRIC BASES OF ROOK GRAPHS

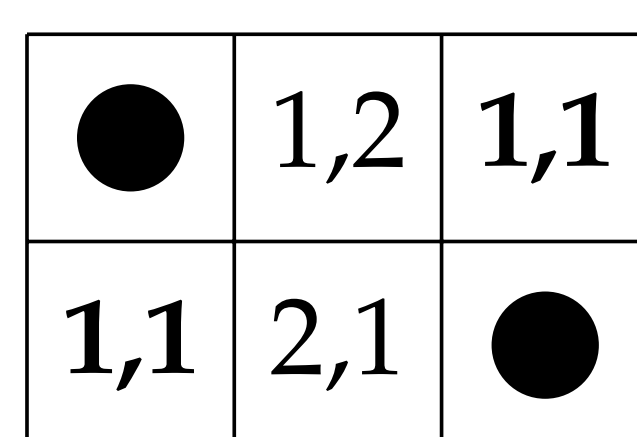
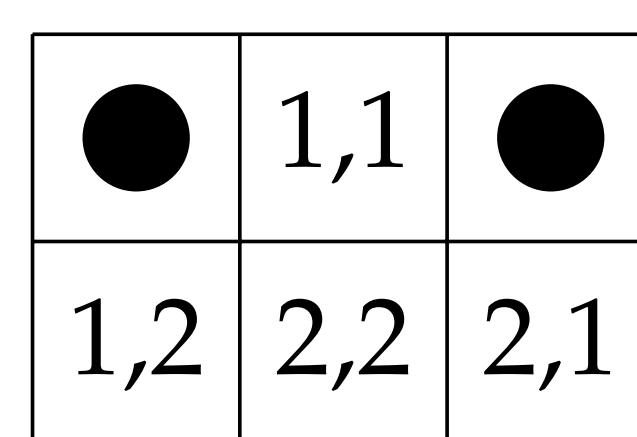
A **complete** graph K_n is a graph where every vertex is connected to every other vertex. A **rook** graph $K_n \square K_m$ is the **Cartesian product** of two complete graphs.

Ex. A rook graph $K_2 \square K_3$



A **token** is a vertex in a metric basis.

Ex. A valid (left) and invalid (right) basis



A token is **lonely** if it is the only one in its row and column.

Ex. The (right) basis has two lonely tokens.

REFERENCES

- [1] J. Cáceres et al., "On the metric dimension of Cartesian products of graphs," *SIAM J. Discrete Math.*, vol. 21, no. 2, pp. 423–441, 2007.
- [2] N. Nishimura, "Introduction to reconfiguration," *Algorithms (Basel)*, vol. 11, no. 4, Paper No. 52, 25, 2018.

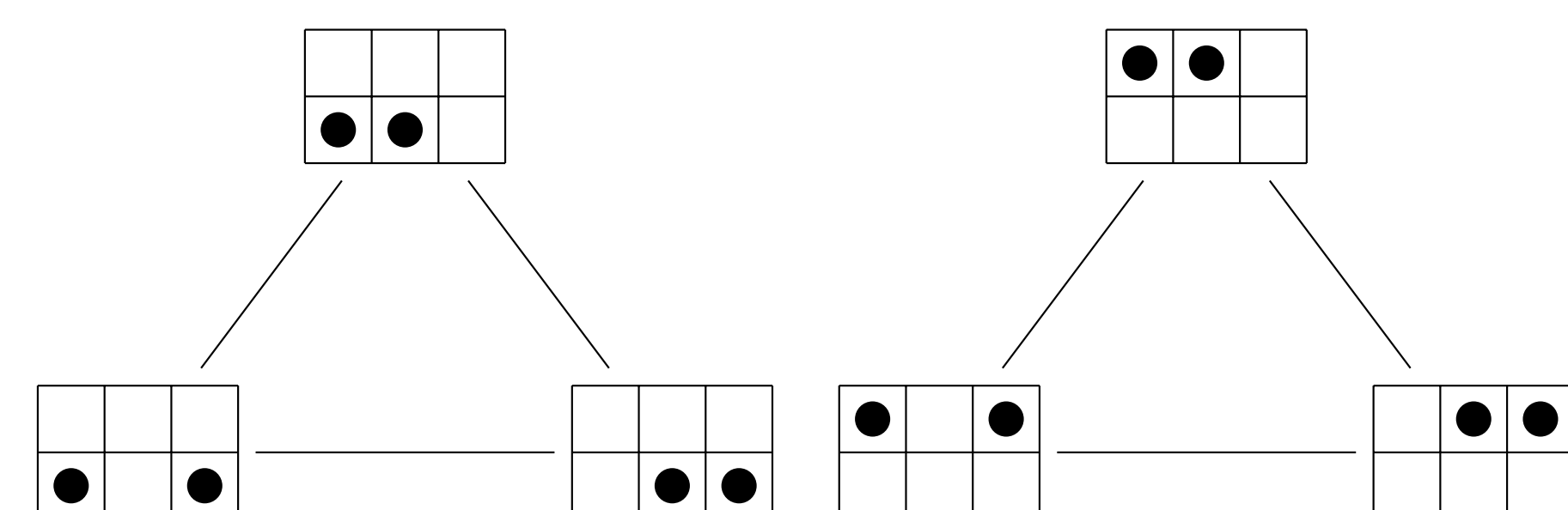
TOKEN SLIDING

Lemma (Cáceres et al., 2007). For $m, n \geq 2$, a set resolves $K_m \square K_n$ if and only if:

1. there is at most one empty row and at most one empty column
2. there is at most one **lonely** token
3. if there is an empty column and empty row, there is no lonely token

A token can **slide** along an edge to a new position in the same row or column, if the resulting set is a **basis**.

Ex. The slide graph $\text{Slide}(K_2 \square K_3)$



QUESTION

Can we transform a basis into any other basis by a sequence of slides?

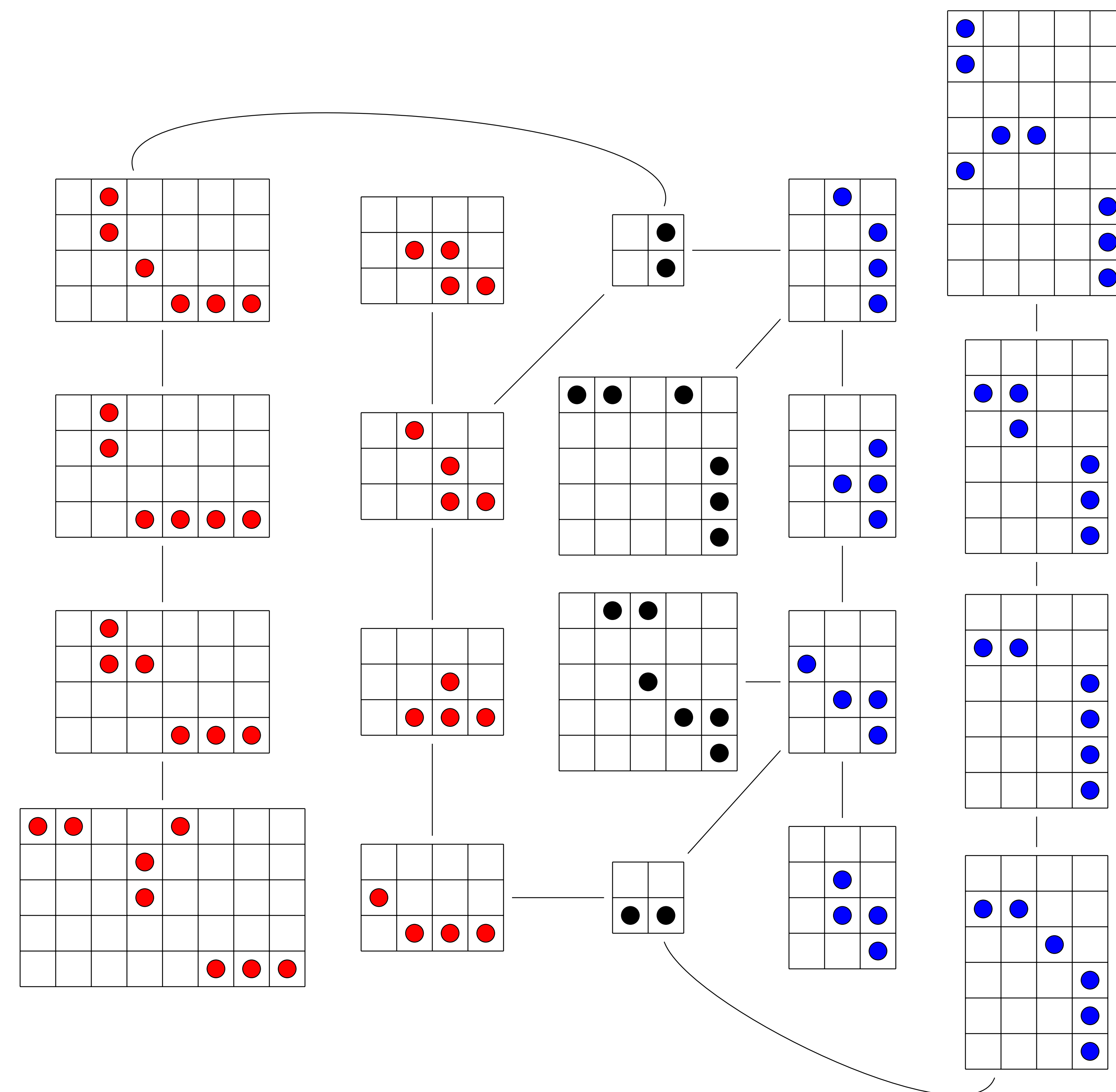
RESULTS

Theorem (C-F-M, 2026). $\text{Slide}(K_m \square K_n)$ is disconnected if and only if $m = n = 2$ or $m = 2x + y + 1, n = x + 2y + 1$ for $x, y \geq 0$.

m, n	2	3	4	5	6	7	8
2	X	X	✓	✓	✓	✓	✓
3		✓	✓	X	✓	✓	✓
4			X	✓	✓	X	✓
5				✓	X	✓	✓
6					✓	✓	X
7						X	✓
8							✓

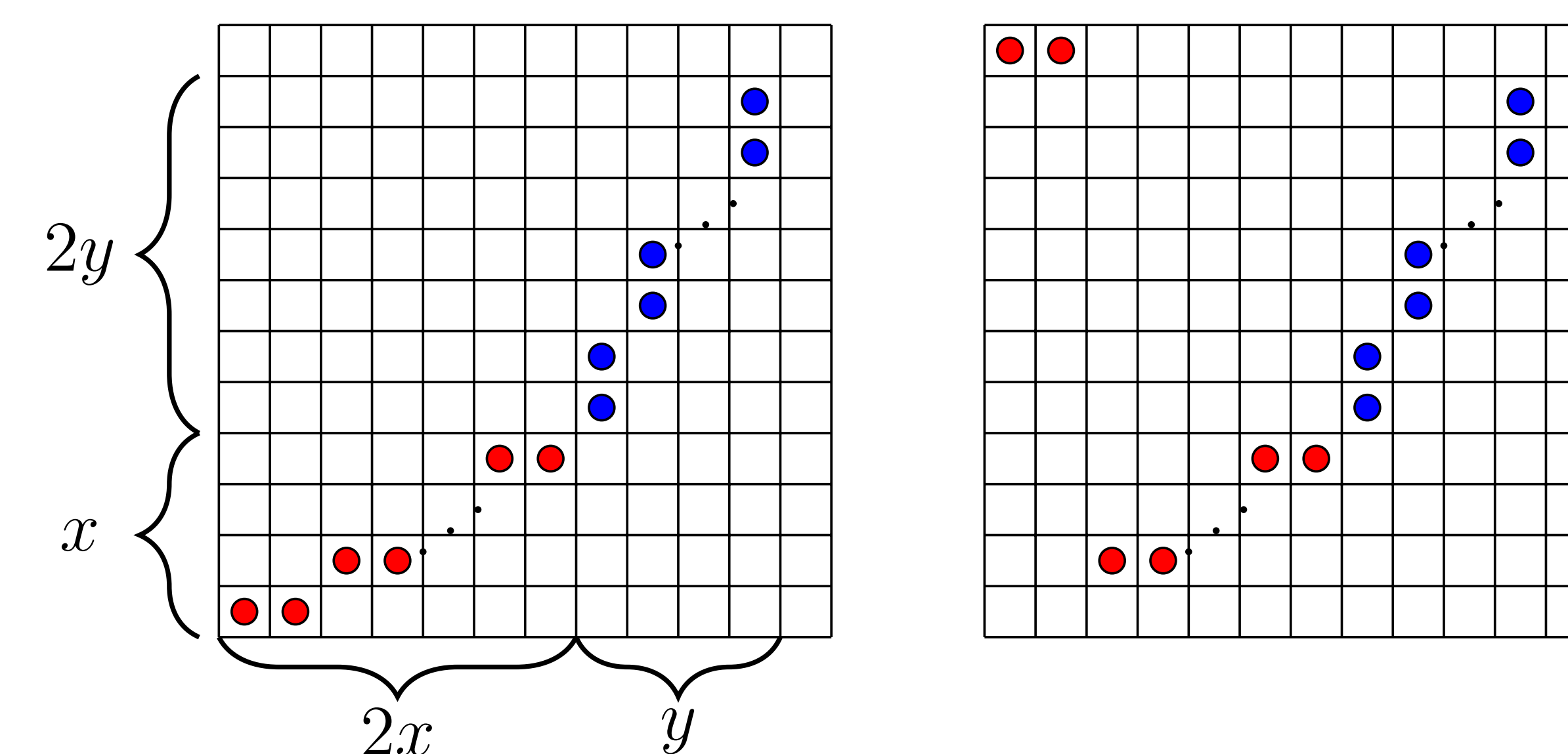
Connectivity of $\text{Slide}(K_m \square K_n)$

MOST SLIDE GRAPHS ARE CONNECTED ...



Connections between representatives for bases of $K_{2x+y+2} \square K_{x+2y+2}$

HOWEVER, THERE IS ONE DISCONNECTED CASE



Two bases that are disconnected on a $K_{2x+y+1} \square K_{x+2y+1}$ graph

FUTURE RESEARCH

- Shapes of reconfiguration graphs
- Algorithmic questions (shortest path)
- Other graph families (Wheels, Planar)

ACKNOWLEDGEMENTS

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